

The ARC Principle: a cross-domain map of its mathematical foundations

The formula $\alpha = d/(d+1)$ already appears independently in two major research programs — West-Brown-Enquist metabolic scaling and Bettencourt's urban scaling theory — lending the ARC Principle's core equation genuine empirical resonance, even though the framework itself remains unpublished, unreviewed, and uncited. The underlying mathematical claim — that Cauchy's functional equations constrain all well-behaved recursive composition to exactly three forms (power law, exponential, logarithmic/saturating) — is a theorem, not a conjecture. What remains unproven is the specific bridge claim: that this constraint governs "recursive amplification" broadly enough to unify scaling phenomena across physics, biology, economics, neuroscience, and AI. This report maps every connection the framework touches, identifies where the three-form constraint and the dimensional formula already live in established science, and flags where the framework makes novel but untested predictions.

The framework's origin and academic standing

The ARC Principle was published by Michael Darius Eastwood in early 2026 as three OSF Preprints ([GitHub](#)) and a book titled *Infinite Architects*. OSF Preprints is an open-upload platform with moderation for spam but **no peer review**. The framework has **zero academic citations**, zero independent replications, and zero engagement on its associated GitHub repository. Eastwood's public profiles describe him as an "AI Expert | Litigant | Creative Director | Producer" ([Instagram](#)) — a distinct figure from the well-known mathematician Michael G. Eastwood (Adelaide/Princeton, conformal geometry). The framework does include explicit falsification criteria, ([GitHub](#)) which is methodologically commendable, but as of March 2026 no external researcher has tested them.

The core equation is $U = I \times R^\alpha$, ([GitHub](#)) where U is effective capability, I is base potential, R is recursive depth, and $\alpha = d/(d+1)$ with d representing effective system dimensionality. ([GitHub](#)) The derivation reportedly proceeds from Cauchy's functional equations (1821): since any measurable function satisfying $f(x \cdot y) = f(x) \cdot f(y)$ must be a power law x^c , ([Imomath](#)) any measurable function satisfying $f(x+y) = f(x) \cdot f(y)$ must be exponential e^{cx} , and any measurable function satisfying $f(x+y) = f(x) + f(y)$ must be logarithmic $c \cdot \log(x)$, ([Imomath](#)) recursive composition operators are restricted to these three families. The specific formula $\alpha = d/(d+1)$ is then claimed to emerge from space-filling optimization in d dimensions.

The mathematical foundation is solid. The claim of universal applicability is ambitious and unverified.

Cauchy's equations genuinely constrain scaling to three forms

The mathematical bedrock of the ARC Principle is not speculative — it is classical analysis. Cauchy proved in his 1821 *Cours d'analyse* that every continuous solution of $f(x+y) = f(x) + f(y)$ is linear: $f(x) = cx$. ([Wolfram MathWorld](#)) All four variants of the functional equation (additive, multiplicative-additive, logarithmic, power) reduce to this base case through substitution, yielding exactly the linear/power-law, exponential, and logarithmic families. ([Korean Studies Institute](#)) Darboux (1875) weakened continuity to continuity at a single point. ([Wolfram MathWorld](#)) Banach and Sierpiński (1920) further weakened it to **Lebesgue measurability** — the mildest condition satisfied by any physically realizable quantity.

([Korean Studies Institute](#)) Georg Hamel (1905) showed ([StudyLib](#)) pathological non-measurable solutions exist ([Wolfram MathWorld](#)) using the Axiom of Choice, but these are dense in the plane, everywhere discontinuous, and physically unrealizable. ([Wikipedia](#))

Karamata's regular variation theory (1930) extends this. Any measurable function whose asymptotic scaling ratio $f(\lambda x)/f(x)$ converges must converge to λ^ρ — a power law. The representation theorem decomposes all regularly varying functions as $x^\rho \cdot L(x)$, where $L(x)$ is slowly varying (growing no faster than any power). ([Wikipedia](#)) Bingham and Ostaszewski (2015) explicitly state: "*The Cauchy functional equation is not only the most important single functional equation, it is also central to regular variation.*" ([LSE Research Online](#))

The three-form constraint is therefore a **theorem**: under any regularity condition that a physical system satisfies, the only solutions to Cauchy's equations are power laws, exponentials, and logarithms. The framework's mathematical foundation stands on firm ground. The open question is whether "recursive amplification" properly maps onto these functional equations — whether the composition of recursive operations in real systems truly satisfies the axioms of Cauchy's equations.

Metabolic scaling delivers $\alpha = d/(d+1)$ independently

The most striking independent validation of the ARC Principle's dimensional formula comes from the West-Brown-Enquist (WBE) metabolic scaling theory. In their landmark 1997 *Science* paper, West, Brown, and Enquist derived that metabolic rate B scales as $B \propto M^{3/4}$ from three axioms: resource distribution networks are space-filling, terminal units (capillaries) are size-invariant, and natural selection minimizes energy dissipation. The result $3/4$ is precisely $d/(d+1)$ for $d = 3$ — the spatial dimensionality of organisms.

WBE explicitly acknowledged this dimensional interpretation. In their 2005 *Journal of Experimental Biology* review, they wrote that fractal-like distribution networks confer "an additional effective fourth dimension" — meaning the hierarchical branching network adds one effective dimension to the three spatial dimensions, and the metabolic exponent is the ratio of the two: $3/(3+1)$. All quarter-power biological scaling laws — heart rate $\propto M^{-1/4}$, lifespan $\propto M^{1/4}$, circulation time $\propto M^{1/4}$ — follow as multiples of $1/(d+1) = 1/4$.

The empirical picture is nuanced. Kleiber's original 1932 measurement gave an exponent of **0.74**, rounded to $3/4$ for slide-rule convenience. Modern analyses across mammals generally find exponents between **0.72 and 0.76** for basal metabolic rate, though Dodds, Rothman, and Weitz (2001) showed that for small mammals the exponent is closer to $2/3$ — which equals $d/(d+1)$ for $d = 2$, corresponding to surface-to-volume scaling. Kolokotronis et al. (2010, *Nature*) revealed that the log-log relationship is actually **convex**, not a pure power law, with different exponents dominating at different mass ranges. Glazier's conditional metabolic theory (2005) argues the exponent varies between $2/3$ and 1 depending on metabolic state and activity level.

This debate can be reframed within the $d/(d+1)$ framework: the effective dimensionality d is not fixed but varies with organism type, activity level, and the degree to which fractal branching networks dominate resource distribution. Surface-dominated scaling gives $d = 2$ ($\alpha = 2/3$); network-dominated scaling gives $d = 3$ ($\alpha = 3/4$); volume-filling without fractal networks gives $d \rightarrow \infty$ ($\alpha \rightarrow 1$, isometric scaling). The framework thus does not predict a single exponent but a family parameterized by effective dimensionality — which is empirically testable.

Urban scaling recapitulates the same formula in social systems

Bettencourt's 2013 *Science* paper derives urban scaling exponents from first principles using what amounts to the same dimensional argument. His framework yields all exponents from two parameters: spatial dimension D and fractal dimension of human travel paths H . ^(uchicago) The key formula is $\alpha = D/(D+H)$, which reduces to $d/(d+1)$ when $H = 1$ (linear exploration paths). For cities on a 2D surface with $H = 1$: $\alpha = 2/3$, predicting infrastructure area scales as population $^{2/3}$, socioeconomic output scales superlinearly as $N^{1+1/6} \approx N^{1.17}$, and per-capita infrastructure costs decline as $N^{-1/6}$. ^(uchicago) Measured exponents — **0.85 ± 0.04** for infrastructure, **1.13 ± 0.02** for GDP — align reasonably well, though not perfectly, with these predictions. ^(University of Chicago)

The parallel to WBE is structural: both derive power-law scaling from the interplay between the spatial dimensionality of the system and an additional dimension introduced by hierarchical or fractal organization. ^(arXiv) The WBE result ($3/4$ from $d=3$ with $+1$ from fractal branching) and Bettencourt's result ($2/3$ from $D=2$ with $H=1$ from travel geometry) are both instances of the same functional form. Neither author references Cauchy's functional equations or a unified "recursive amplification" principle, but both independently arrive at the same mathematical structure.

Renormalization group theory produces the same three forms from different axioms

The renormalization group (RG) in physics provides the deepest independent parallel to the three-form constraint. RG transformations form a semigroup under composition — coarse-graining a system at scale λ_1 and then at λ_2 is equivalent to coarse-graining at $\lambda_1\lambda_2$. This is precisely the structure captured by Cauchy's multiplicative functional equation $f(xy) = f(x) \cdot f(y)$, whose solutions are power laws.

Near RG fixed points, physical observables exhibit exactly the three Cauchy-constrained scaling forms: **power laws** at critical fixed points (correlation functions decay as $r^{-(d+2-\eta)}$), **exponential** decay/growth for relevant and irrelevant operators away from fixed points (correlation length diverges as $|T-T_c|^{(-\nu)}$), and **logarithmic corrections** at marginal dimensions (the upper critical dimension $d_c = 4$

for the Ising model, where the coupling runs logarithmically). Kenneth Wilson (Wikipedia) received the 1982 Nobel Prize for demonstrating that universality classes (Wikipedia) — the groupings of systems with identical critical exponents — are determined by just three factors: spatial dimensionality d , order parameter symmetry, and interaction range.

The hyperscaling relation $\nu d = 2 - \alpha$ (where α here is the specific heat exponent, not the ARC exponent) directly ties critical exponents to spatial dimension. (Wikipedia) However, the specific formula $d/(d+1)$ does not appear as any standard critical exponent. The closest structural parallel is the dimensional crossover implicit in the **holographic principle**: information in a d -dimensional volume is bounded by its $(d-1)$ -dimensional boundary, giving a ratio $(d-1)/d$. This differs from $d/(d+1)$ — the holographic formula is $1 - 1/d$ while the ARC formula is $1 - 1/(d+1)$ — but both encode dimensional reduction where a system's effective scaling is less than its nominal dimensionality.

Neural avalanches and the criticality hypothesis

Beggs and Plenz's 2003 discovery that cortical networks produce neuronal avalanches with power-law size distributions (PubMed) (Journal of Neuroscience) (exponent $-3/2$) provides a neuroscience connection. The exponent $3/2$ is the mean-field prediction for a critical branching process, arising when the branching parameter $\sigma = 1$ (Scholarpedia) (each spike triggers on average one downstream spike). Intriguingly, $3/2 = d/(d+1)$ for $d = 3$, matching the spatial dimensionality of cortical tissue — though this numerical coincidence has not been explored in the avalanche literature.

The brain's apparent self-organization toward criticality ($\sigma \approx 1$) supports the broader claim that recursive amplification processes are attracted to critical scaling. At criticality, neural networks maximize dynamic range, information transmission, and sensitivity to inputs. Deviations from criticality — subcritical ($\sigma < 1$, exponentially damped activity) or supercritical ($\sigma > 1$, runaway excitation/seizure) — map onto the exponential and saturating Cauchy forms. The critical point itself produces the power-law form.

Neural scaling across species adds another dimension. Herculano-Houzel's neuron-counting work reveals that **primate brains scale nearly linearly** (mass $\propto$ neurons $^{1.0}$) while **rodent brains scale hypermetrically** (mass $\propto$ neurons $^{1.7}$). Primate brains achieve more neurons per gram of tissue, enabling denser recursive circuitry — consistent with the idea that architectural efficiency determines how recursive amplification translates to cognitive capability.

Information theory and AI scaling laws

Shannon's channel capacity theorem $C = B \cdot \log_2(1 + S/N)$ demonstrates that information throughput scales logarithmically with signal power (Wikipedia) — one of the three Cauchy forms (saturating). Landauer's principle sets an absolute floor of $k_B T \cdot \ln(2)$ energy per bit erased, (PubMed Central) creating a fundamental resource constraint on any recursive computation. Both exemplify how physical bounds force scaling into specific functional forms.

Neural scaling laws in AI provide the most data-rich modern testing ground. Kaplan et al. (2020) found that language model loss decreases as a **power law** of model parameters: $L(N) \propto N^{-0.076}$. (Medium) (Emergent Mind) Hoffmann et al. (2022, "Chinchilla") revised this but maintained the power-law form. (Medium) (arXiv) Crucially, Sharma and Kaplan (2022) proposed that the scaling exponent depends on the **intrinsic dimension of the data manifold**: $\alpha \approx 4/d_{\text{manifold}}$. Bahri et al. (2024, PNAS) showed that in the resolution-limited regime, exponents depend on the data manifold dimension d and the kernel spectrum. (PNAS) A December 2024 paper used percolation theory to derive power-law scaling from criticality in dataset structure.

Test-time compute scaling (2024–2026) reveals **logarithmic/saturating** returns when measured as accuracy gains per additional compute — achieving linear accuracy improvement requires exponential compute increases. (Substack) Chen et al. (2024) proved that different test-time strategies produce either power-law or exponential decay of failure probability, directly instantiating two of the three Cauchy forms. (arXiv) The third (saturating) appears as the practical ceiling: inference costs grow unboundedly for diminishing gains.

The relationship between Karamata's theory and AI scaling is underexplored. (Encyclopedia of Mathematics) If neural scaling exponents are determined by data manifold dimension (as Sharma and Kaplan suggest), and if Karamata's theorem constrains all scaling to power-law forms with slowly varying corrections, then the power-law nature of neural scaling laws may be a mathematical necessity rather than an empirical surprise.

Recursive amplification in consciousness and alignment

Theories of consciousness converge on recursive processing as essential. Lamme's recurrent processing theory posits that feedforward sweeps (~100ms) support unconscious categorization, local recurrence (~200ms) enables perceptual organization, and global recurrence (~300ms+) generates reportable conscious experience — a hierarchy where **recursive depth determines the level of consciousness**. The Global Neuronal Workspace Theory models conscious ignition as a nonlinear phase transition ([PubMed Central](#)) from local to global recurrent activity, exhibiting threshold dynamics consistent with constrained amplification. The COGITATE adversarial collaboration (published *Nature*, 2025) found that neither IIT nor GNWT was fully supported, with conscious content distributed across visual, ventrotemporal, and inferior frontal cortex — suggesting recursive processing operates at multiple scales simultaneously.

In AI alignment, recursive amplification is both the goal and the problem. Recursive reward modeling (Leike et al., 2018) proposes helper agents that recursively assist humans in evaluation. AI safety via debate (Irving et al., 2018) achieves **PSPACE** amplification of polynomial-time judges through recursive argumentation ([arXiv](#)) — exponential capability gain through logarithmic-depth game trees. Constitutional AI (Bai et al., 2022) implements recursive self-correction through iterated critique-revision cycles, ([arXiv](#)) ([Anthropic](#)) which empirically show **saturating** improvement after several iterations. Weak-to-strong generalization (Burns et al., 2023) found that strong models supervised by weak ones outperform their supervisors but don't fully recover ceiling performance ([ACM Digital Library](#)) ([arXiv](#)) — partial recovery consistent with $d/(d+1) < 1$ efficiency per recursive step. A 2025 *Scientific Reports* paper proved via Rice's theorem that inner alignment is **undecidable** for arbitrary architectures, arguing alignment must be a guaranteed architectural property ([Nature](#)) — resonating with the ARC Principle's claim that scaling behavior is architecture-dependent.

Where the framework generates novel predictions

Across all seven domains, the ARC Principle's contact with established science falls into three categories:

The **strongest connections** are to WBE metabolic scaling and Bettencourt's urban scaling, where $d/(d+1)$ appears independently with the same interpretation — effective dimensionality determines the scaling exponent. The three-form constraint from Cauchy's equations is mathematically rigorous and implicitly present in RG theory, statistical physics, and information theory. Neural avalanche exponents and AI scaling laws are consistent with the framework.

The **novel predictions** that would distinguish the ARC Principle from existing theories include: (1) that test-time compute scaling in AI should follow power-law form with exponents predictable from the effective dimensionality of the reasoning task; (2) that alignment difficulty should scale as a specific function of recursive depth, with the exponent determined by the architecture's effective dimensionality; (3) that assembly theory's molecular complexity index should be expressible as a logarithmic (saturating) Cauchy form; and (4) that Φ in IIT should scale with system size according to a $d/(d+1)$ power law once architecture is controlled for.

The **weakest connections** are to constructor theory, loop quantum gravity, and Orch-OR, where the framework makes no specific testable predictions beyond broad conceptual resonance.

Conclusion: genuine mathematical bones, unproven empirical body

The ARC Principle rests on an **authentic mathematical theorem** (Cauchy's three-form constraint) and connects to an **independently discovered dimensional formula** ($d/(d+1)$ in WBE and Bettencourt scaling). These are not trivial coincidences — they reflect the deep mathematical fact that space-filling optimization in d dimensions with hierarchical branching yields the same functional form regardless of whether the system is a vascular network, a city, or potentially a neural architecture.

What remains undemonstrated is the universal bridge: that "recursive amplification" as defined by the ARC Principle correctly formalizes the compositional structure of these diverse systems such that Cauchy's constraint applies. The renormalization group achieves something similar for phase transitions through the semigroup property of coarse-graining, but that derivation is specific and rigorous — not a universal assertion. The most productive next steps would be **domain-specific falsification tests**: measuring whether AI test-time scaling exponents correlate with task dimensionality as the framework

predicts, or whether neural scaling across architectures follows $d/(d+1)$ with d estimated from network topology. Until such tests are conducted and published in peer-reviewed venues, the ARC Principle remains a suggestive but unvalidated conjecture — one with better mathematical foundations than most such frameworks, but whose extraordinary scope demands extraordinary evidence.